

KONYA GIDA VE TARIM ÜNİVERSİTESİ

Faculty of Engineering and Architecture

Computer Engineering

Digital Design

Digital Traffic Light Controller Design

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1-INTRODUCTION:

In this project, a digital traffic light control system for a four-way intersection has been designed. The system utilizes sequential and combinational logic circuits to manage the timing and sequence of traffic signals (Red, Yellow, Green), ensuring safe and efficient traffic flow while preventing signal conflicts.

Project Scope:

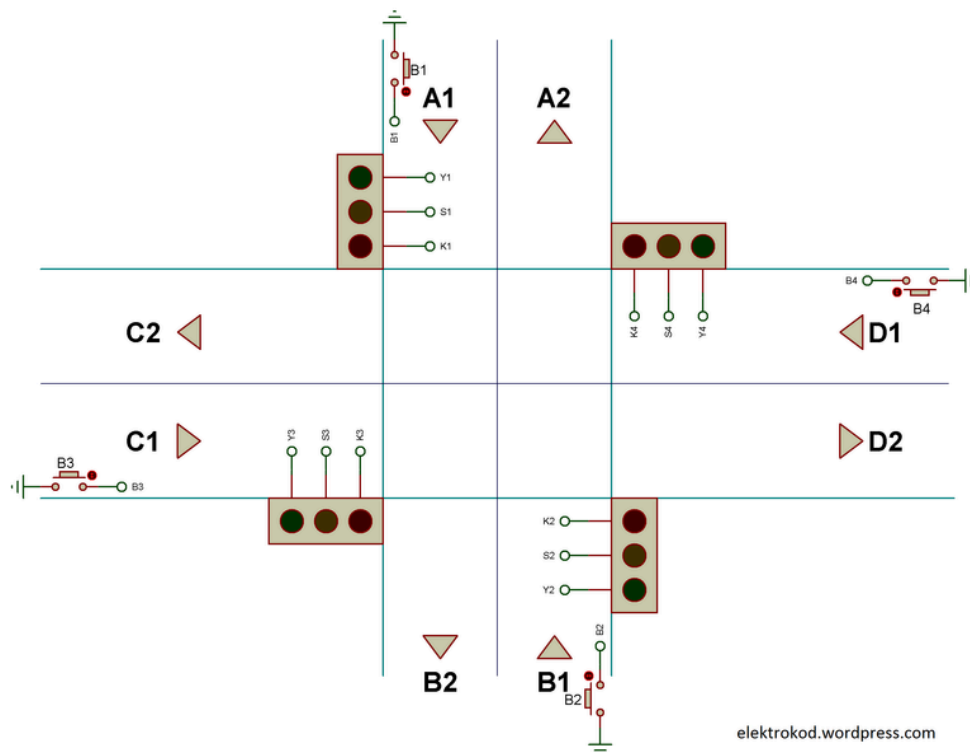
- Design of a Finite State Machine (FSM) for traffic logic
- Implementation of sequential circuits using 7400-series ICs (7474 Flip-Flops, Logic Gates)
- Timing control and state transition logic
- Circuit design and simulation with Logisim
- Visualization of 4-way intersection scenarios with LED indicators

This work enables students to better understand digital logic design principles, specifically sequential circuits and state machine implementation, providing hands-on experience with fundamental concepts of computer organization.

2. HARDWARE MODULES

- 7474 IC (Dual D Flip-Flop): Stores the current state bits of the Finite State Machine (FSM).
- 7408 IC (Quad 2-Input AND): Implements combinational logic for state transitions.
- 7486 IC (Quad 2-Input XOR): Used for exclusive-OR logic operations in control circuitry.
- 7404 IC (Hex Inverter): Provides signal inversion (NOT operation) for logic flow.
- Clock Generator: Generates square wave pulses to trigger synchronous state changes.
- LED Indicators: Visual output representing Traffic Lights (Red, Yellow, Green).
- Wiring & Interconnections: Connects logic gates and flip-flops to form the data path.

3. Project Template



4. PROJECT OBJECTIVE AND WORKING PRINCIPLE

4.1. Project Objective: The primary objective of this project is to design and simulate a fully functional traffic light control system for a busy four-way intersection. The main goal is to ensure traffic safety and prevent accidents by strictly timing the red, yellow, and green signals.

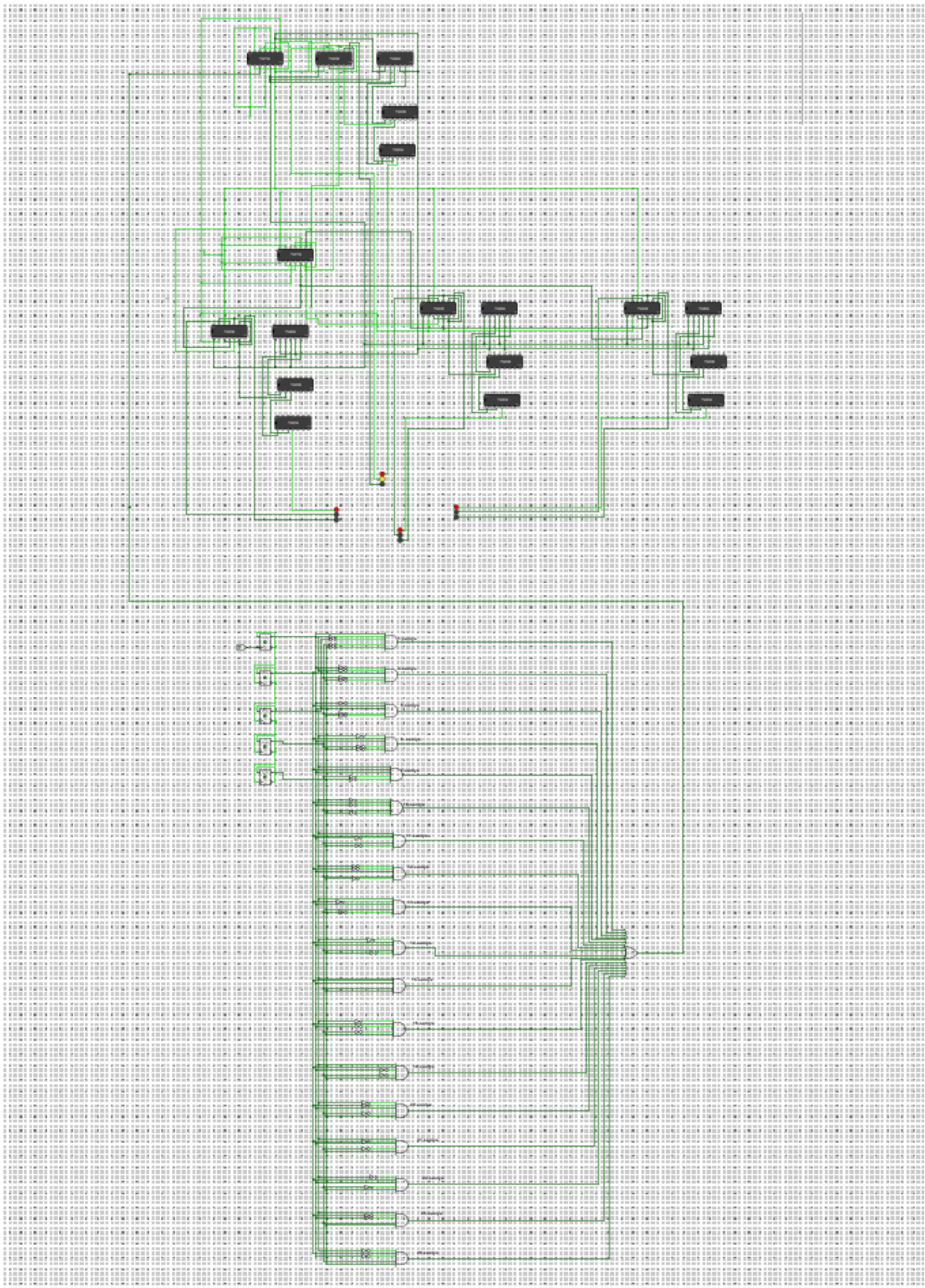
From an educational perspective, this project aims to demonstrate the practical application of digital logic design. It specifically focuses on how Sequential Circuits and Finite State Machines (FSM) operate in real-world scenarios. By using standard components like the 7474 Flip-Flops and logic gates, the project bridges the gap between theoretical Boolean algebra and practical circuit implementation.

4.2. Working Principle: The system operates as a synchronous digital circuit driven by a clock signal. The core working principle is based on a 4-bit Finite State Machine (FSM).

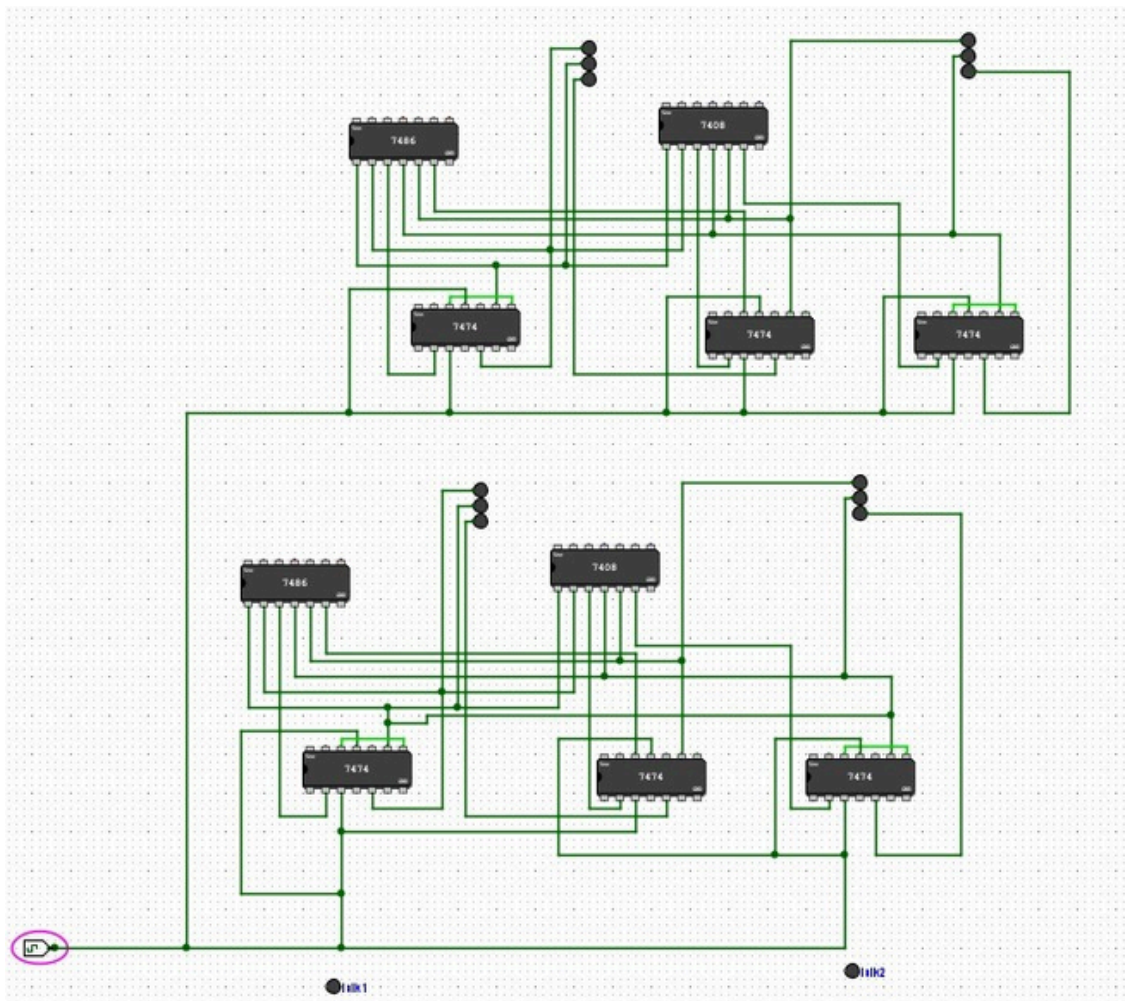
The operation can be described in three main steps:

1. State Storage: The system uses four D Flip-Flops (7474 ICs) to store the current state of the traffic lights. Since there are 4 bits, the system can handle a specific sequence of binary states (e.g., 0000, 0001, etc.).
2. Next State Logic: On every rising edge of the clock pulse, the combinational logic circuit (built with AND, XOR, and NOT gates) calculates the next state based on the current values. This ensures that the lights change in a specific, safe order (Red -> Red/Yellow -> Green -> Yellow -> Red).
3. Output Generation: The outputs of the flip-flops are connected to the LED indicators. When the system reaches a specific binary state, the corresponding LEDs turn on to regulate the traffic flow for the intersection.

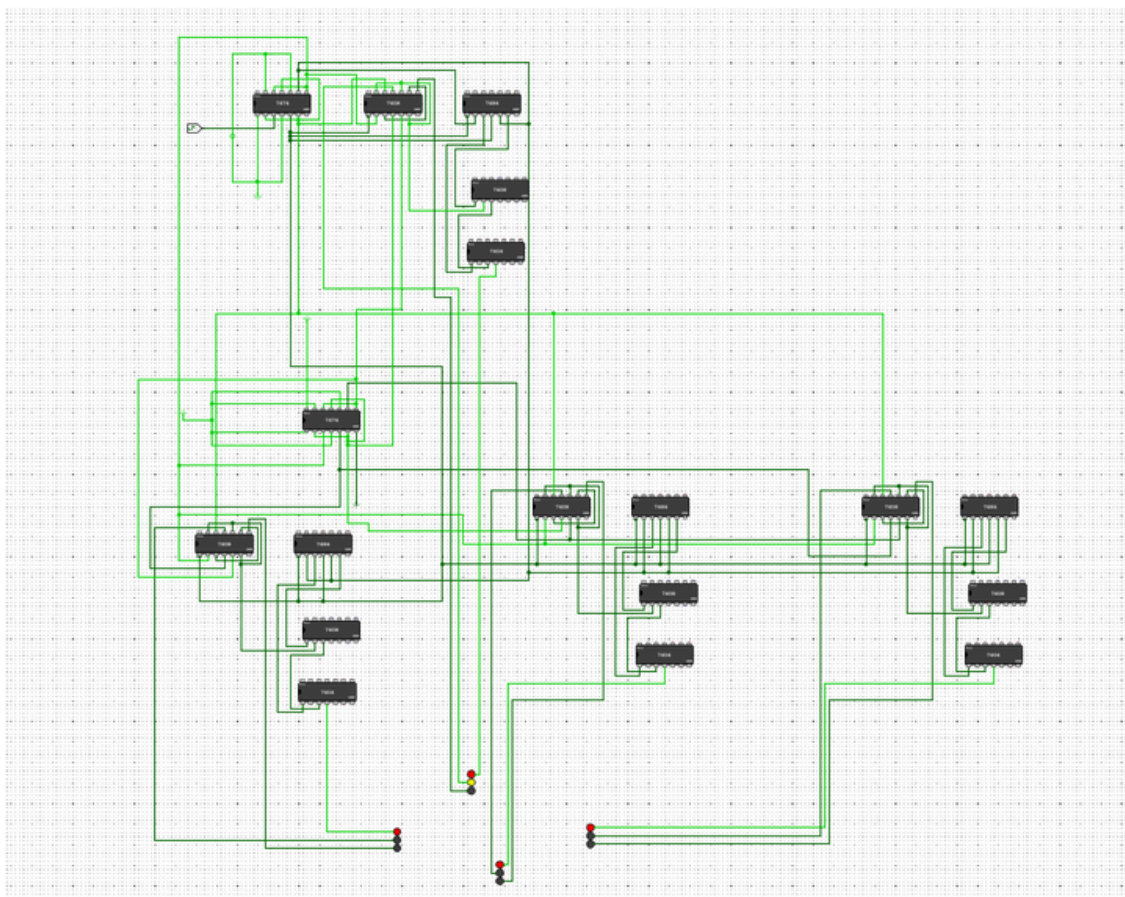
5- Logisim Circuit
Main Circuit



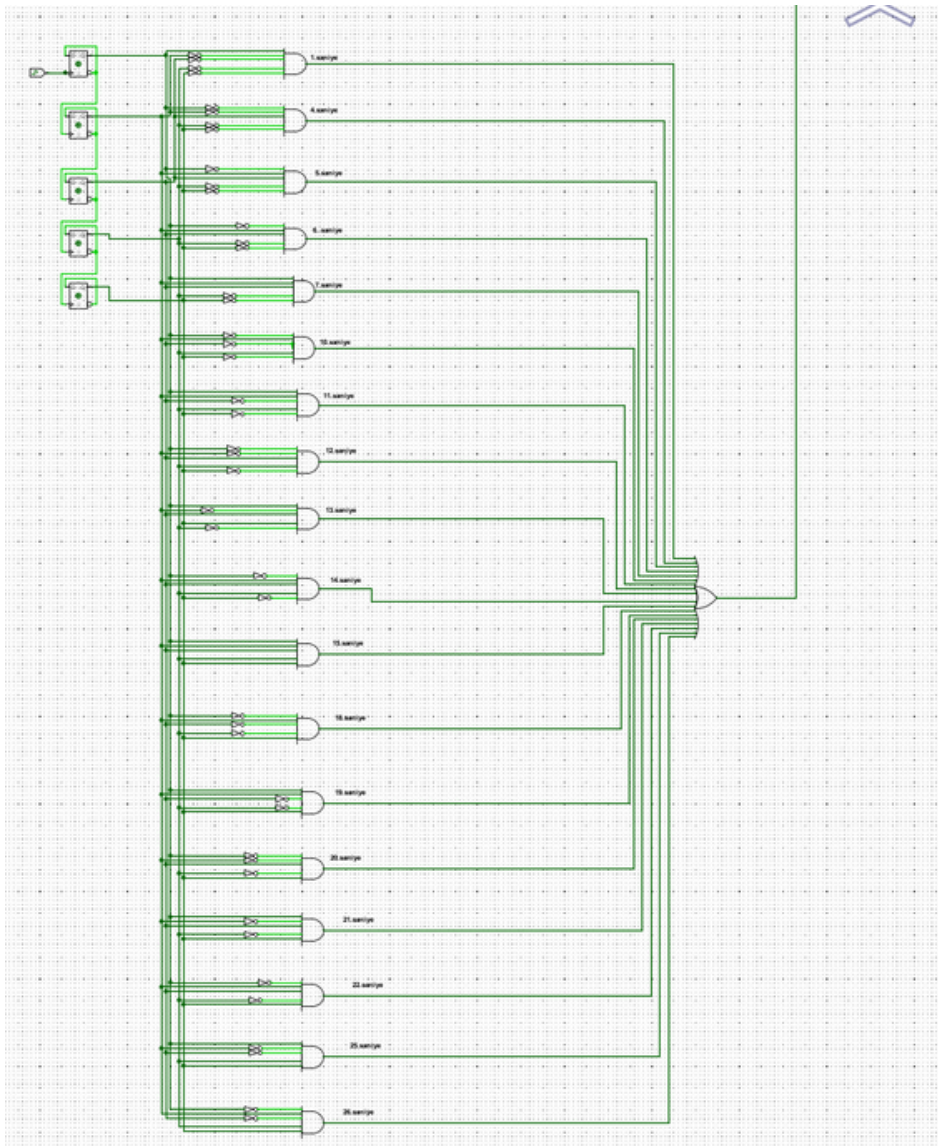
Traffic Light



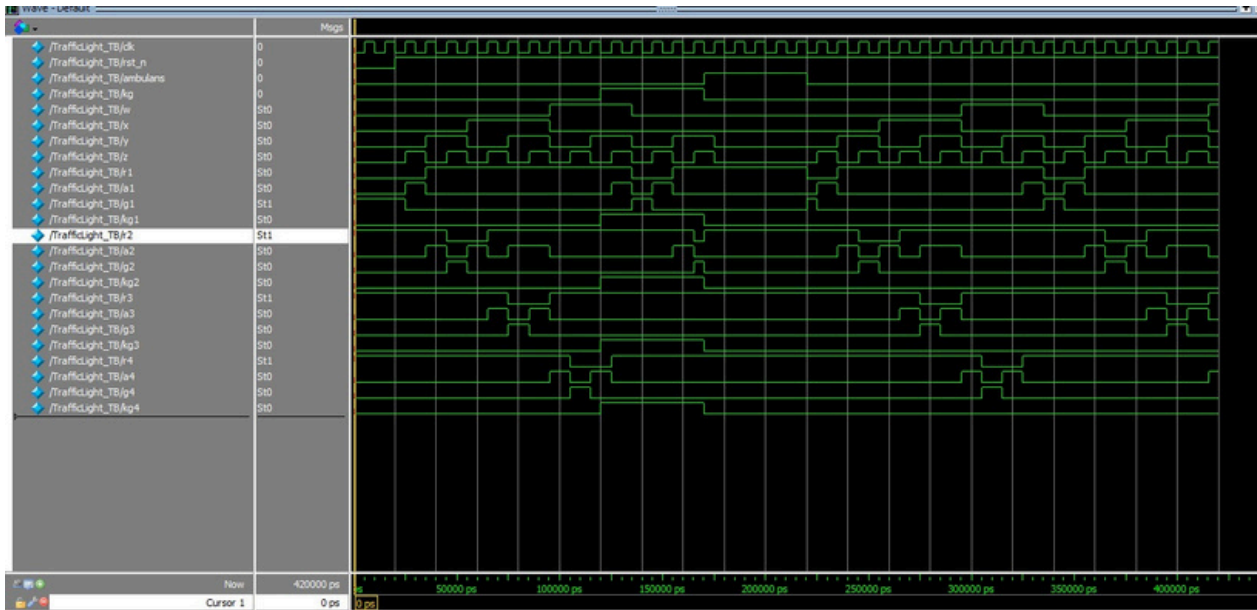
Selector



Timer



Verilog Wawe



Conclusion

In this project, a digital traffic light control system was successfully designed and simulated using Logisim. By employing a Finite State Machine (FSM) architecture with D Flip-Flops and standard logic gates, the circuit effectively manages the traffic flow at a four-way intersection.

The simulation results confirmed that the system transitions correctly between states (Red, Yellow, Green) and operates synchronously with the clock signal. This project demonstrated the practical application of sequential logic circuits and verified that functional control systems can be built using fundamental hardware modules.